

**Impacts of Per- and Polyfluoroalkyl Substance Concentration in Drinking Water on
Cancer Rates**

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Introduction

Per- and polyfluoroalkyl substances (PFAS) are a class of thousands of substances used in consumer and industrial settings. While costly to produce, PFAS have chemical properties that make them useful in various applications (Glüge, et al., 2020). Their durability and persistence are well-known characteristics, but are also a leading cause for concern. When released into water sources (often through industrial runoff or firefighting activities), they accumulate and will stay present for a long time. Traditional water treatment methods do not typically remove PFAS from the water, which means that they may often end up in drinking water (Abunada, Alazaiza, & Bashir, 2020).

Various studies on the impact of PFAS have found sufficient evidence that there is an association between PFAS and decreased antibody response, dyslipidemia (a disorder characterized by abnormally high or low amounts of lipids or lipoproteins in the blood), decreased infant and fetal growth, and an increased risk of kidney cancer (National Academies of Sciences, Engineering, and Medicine, 2022). Additionally, there is some evidence to suggest that there are also associations with other health issues, such as various cancers, thyroid disease, and ulcerative colitis (National Academies of Sciences, Engineering, and Medicine, 2022).

Some studies have aimed to measure potential impacts of high levels of exposure to PFAS following water contamination events, such as a study that analyzed the effects of such an event in Merrimack, New Hampshire, which found associations with multiple elevated levels of health concerns (Panikkar, et al., 2019). Additionally, research has found that the amount of PFAS in blood plasma that comes from water as a percent of all PFAS in people's plasma is significant: around 12% for PFOA and 13% for PFNA, 2 major types of PFAS (Hu, et al., 2019). Despite these findings, there appears to be a lack of significant academic research on the impact of general PFAS concentrations in drinking water on human health.

This paper will look at 3 types of PFAS: PFOA, PFNA, and PFOS. As mentioned above, PFOA and PFNA are 2 major types of PFAS in tap water that have been found in blood plasma concentrations (Hu, et al., 2019). PFOA as well as PFOS are generally some of the most well-studied types of PFAS due to their long-lasting occurrence in the environment and associated adverse health effects (Habib, et al., 2024). This paper aims to explore whether levels of these 3 types of PFAS in drinking water are significantly associated with non-skin cancer prevalence in the United States. The null hypothesis is that there is no significant association between high levels of PFOA, PFNA, and PFOS on cancer prevalence. The alternative hypothesis is that there is a significant association between high levels of these PFAS on cancer prevalence.

Two different datasets were used to do this analysis. The first dataset is from Occurrence Data from the Unregulated Contaminant Monitoring Rule (UCMR) provided by the Environmental Protection Agency (EPA). The UCMR is used by the EPA to collect data for contaminants in drinking water that do not actually have regulatory standards set under the Safe Drinking Water Act. The UCMR 5, which is the dataset that is used in this paper, specifically looked at occurrence levels of 29 types of PFAS (including PFOA, PFNA, and PFOS) in public

water systems (PWSs), including the zip codes that those PWSs serve (Environmental Protection Agency, 2025).

The second dataset is the 2024 PLACES data, provided by the Centers for Disease Control and Prevention (CDC). As stated on the CDC's website: "PLACES provides model-based estimates based on data from Behavioral Risk Factor Surveillance System (BRFSS), Census decennial population counts and annual county population estimates, and the American Community Survey 5-year estimates" (Centers for Disease Control and Prevention, 2024). The PLACES data measures the approximate prevalence of various health and health-related categories by zip code, as a percentage of the total population of that zip code. Because zip code was available in both of these datasets, I decided to do my analysis at the zip code level. Additionally, zip code provides a level of granularity that may reveal more community-specific effects of higher levels of PFAS that would not be apparent at the county or state level. Another important note about this dataset is that it only includes the population of each zip code that is 18 years or older, and all prevalence rates are based on the adult population only.

Methods

The response variable explored in this paper was non-skin cancer prevalence as a percent of the population of a zip code. This is because various types of cancer have been cited as a common adverse health impact of long-term PFAS exposure. Other health outcomes measured in the PLACES data were either more vague or not a known health effect of PFAS exposure.

The primary explanatory variables were PFAS concentration in $\mu\text{g/L}$, with levels of PFOA, PFNA, and PFOS measured as separate variables. One thing to note about this variable is that the levels of these PFAS can only be detected above a certain level, known as the minimum reporting level (MRL), which is "the quantitation limit for a contaminant that is considered achievable, with 95% confidence, by at least 75% of laboratories nationwide using a specified analytical method" (Environmental Protection Agency, 2025). For this paper, if a PFAS was reported as less than the MRL, the MRL was used as the level for that zip code.

Other explanatory variables that were used to account for variation in cancer rates were food insecurity prevalence, smoking prevalence, percentage of the zip code population that had a routine checkup in the last year, and total population of the zip code that is 18 years or older (because the CDC dataset only measures the adult population).

Food insecurity prevalence was chosen because it acts as an indicator of socioeconomic status and because people experiencing food insecurity may get screened for cancer at lower rates, and thus may show lower rates of cancer overall (Shah, Islam, & Braunstein, 2023). The total population of the zip code older than 18 years old was chosen as an explanatory variable because there is an association of higher rates of cancer in urban areas, but certain cancers are more common in rural areas due to factors such as increased tobacco usage, higher rates of HPV, and differences in screening rates (Zahnd, et al., 2018). Smoking prevalence was chosen because it is a known carcinogen; smokers are twice as likely to die from cancer than non-smokers (Ozlu

& Bülbül, 2005). Finally, routine checkup prevalence is an explanatory variable to help account for cases where cancer may go undetected because people simply do not see a doctor to get tested.

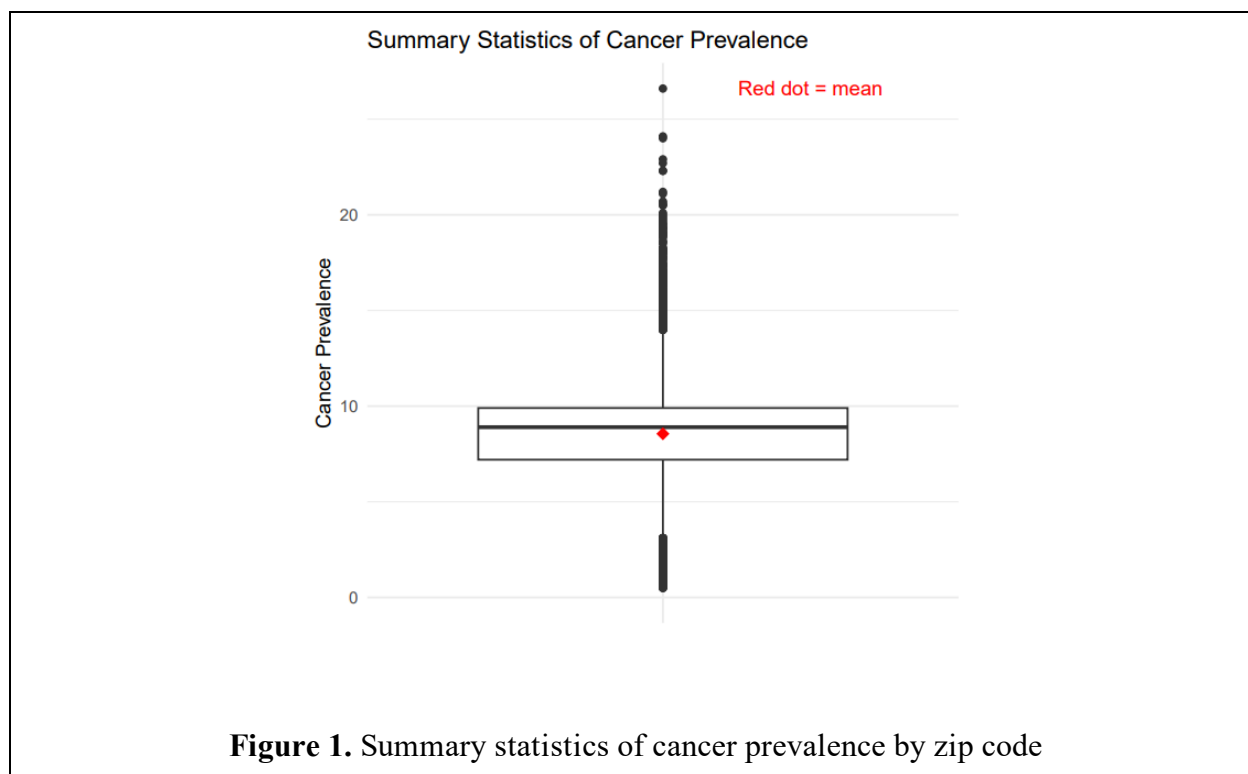
Because the primary explanatory and response variables of this model are both numeric, the most appropriate test for the effects of PFAS concentration on cancer is a linear regression. Before performing a regression, however, it is important to check the assumptions of a linear regression model: linearity, independent observations (given x), normally-distributed errors/residuals, and equal variance for all x 's.

A residual plot was used to check for linearity and equal variance of x 's; however, there was a clear non-linear pattern that was present in the data and not centered around the 0-line, meaning that we cannot assume linearity or equal variance of x 's. A Q-Q plot was used to assess for normally-distributed residuals, but there were large deviations from the normal distribution line, so we cannot assume normally-distributed residuals. A variance inflation factor test was done to check for multicollinearity, and in this case, there were no high VIF values, so we are able to establish that there was no multicollinearity among the predictor variables. Finally, a graph of Cook's distance was done to check for high leverage points, but none were found.

The following R packages were used to do the analysis for this paper: skimr (Waring, et al., 2025), tidyverse (Wickham, Averick, et al., 2019), readr (Wickham, Hester, & Bryan, 2024), and car (Fox & Weisberg, 2019).

Results

Before running the linear regression, we will explore the summary statistics of our response variable, non-skin cancer prevalence. As seen in **Fig. 1**, the mean and median estimated percent of each zip code's population with non-skin cancer are just below 9, at around 8.6 and 8.9, respectively. The inter-quartile range is between 7.2 and 9.9, and the overall range is from a 0.5 minimum value to a 26.6 maximum value. Since there are no groups, this summary provides us with what to expect overall for cancer prevalence in the United States by zip code.



A linear regression model was run with log-transformed PFOA concentration, log-transformed PFNA concentration, log-transformed PFOS concentration, food insecurity prevalence, smoking prevalence, percentage of population with a routine checkup in the last year, and log-transformed total adult population of the zip code as explanatory variables, and Cancer prevalence as the response variable (**Table 1**). The logarithmic values of each of the types of PFAS values were taken because the data were heavily right-skewed, and the logarithmic transformation helps to normalize them. The overall adjusted R^2 of the model is 0.602, meaning that this model explains around 60.2% of the variation in cancer prevalence by zip code. All of the variables (including the intercept) have a p-value less than 0.001, except for the logarithm of PFNA. Interestingly, the 2 values for PFOA and PFOS contradict each other. The logarithm of PFOA has a negative coefficient estimate of -0.802, meaning that a 1% increase in PFOA concentration is correlated with a 0.008 decrease in cancer prevalence. The logarithm of PFOS has a positive coefficient estimate of 0.524, meaning that a 1% increase in PFOS concentration is correlated with a 0.005 increase in cancer prevalence. A graph of the relationship between cancer prevalence and these PFAS *without* the other explanatory variables can be seen in **Fig. 2**, where there is a visible negative correlation between these variables, especially in the case of PFNA. The negative slope is likely due to the exclusion of the other explanatory variables, and is not representative of the full model (**Fig. 2**). A stepwise regression model was run to see if this inverse of slopes could be accounted for, but that model still produced a positive coefficient for PFOS and a negative coefficient for PFOA. The stepwise model only eliminated PFNA as an explanatory variable.

Variable	Coef. Estimate	Standard Error	P-value
Intercept	-13.399***	3.026	3.56E-06
log(PFOA)	-0.802***	0.100	1.25E-15
log(PFOS)	0.505***	0.069	3.65E-13
log(PFNA)	-0.429	0.561	0.444
Food Insecurity Prevalence	-0.176***	0.002	2E-16
log(Total Population 18+)	-0.176***	0.010	2E-16
Current Smoking Prevalence	0.096***	0.004	2E-16
Checkup Past Year Prevalence	0.263***	0.003	2E-16
Adj. R-squared	0.602		
No. Observations	13,392		

Table 1. Tables of results of linear regression where cancer prevalence is the response variable. *** indicates a p-value less than 0.001, and no symbol after the coefficient estimate indicates a p-value greater than 0.1. Numbers in parentheses are the standard errors.

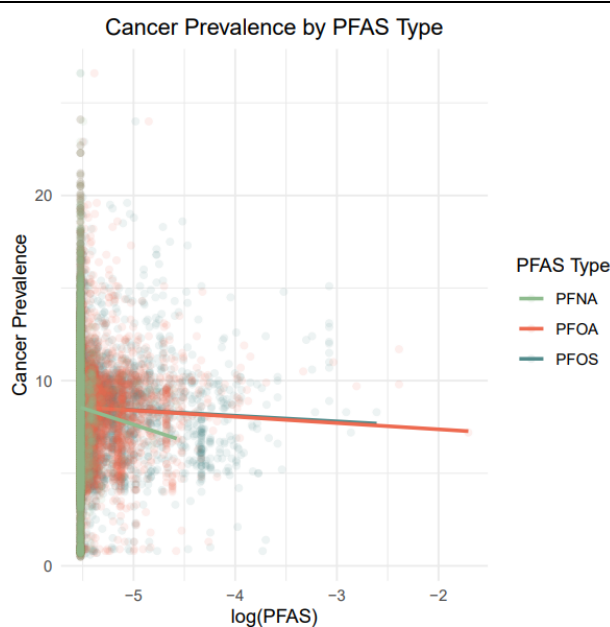


Figure 2. Scatterplot of cancer prevalence versus the logarithmic values of PFNA, PFOA, and PFOS, with a linear regression line for each PFAS shown.

Discussion

Because the linear regression model does not meet the assumptions of a linear regression model—specifically linearity, equal variance of x 's, and normally-distributed residuals—we do not have enough evidence to reject the null hypothesis that there is no significant association between high water concentrations of various types of PFAS and cancer prevalence. In addition to the assumptions not being met, there are conflicting signs to the two statistically significant types of PFAS, PFOA and PFOS, which means that even if the assumptions were met, there

would not be a clear relationship between PFAS concentration in water and cancer prevalence. PFNA did not have a p-value below 0.05, so we cannot reject the null hypothesis for that PFAS either.

Despite its known link to cancers and other diseases, the lack of a significant relationship between PFAS in water and cancer prevalence may be due to a variety of factors. One of these possible explanations is that PFAS exposure from water is a relatively small percentage of the total exposure that the average person has to PFAS. One study found that for PFOS, only 24% of the median daily intake came from water (Lorber & Egeghy, 2011). This could indicate that, outside of cases of extreme PFAS contamination in water, such as in the case of the contamination event in New Hampshire (Panikkar, et al., 2019), the amount of exposure in most water sources is simply not enough to make a significant difference in the likelihood of cancer.

Another possible explanation for the lack of relationship is the large number of observations for PFOA, PFNA, and PFOS that are below the minimum reporting level (MRL). For all three of these types of PFAS, more than 75% of the data is below the MRL, meaning that the data is not only heavily right-skewed, but also almost entirely an estimate. This can be observed quite clearly in **Fig. 2**. Any value that falls below the MRL could be anywhere between 0 µg/L and up to the MRL level, so a majority of the data is based on an unknown number. If we were able to see the actual values of these observations under the MRL, it may be possible to establish a more linear relationship that isn't so centered on the single point that is the MRL.

The final possible explanation for a lack of relationship between the explanatory and response variables may also be the inability, in this case, to measure the variables over time. Unfortunately, the UCMR 5, in which PFAS data specifically were analyzed, only collected data from 2023 to 2025 (Environmental Protection Agency, 2025), and the most recent PLACES data (2024 release) was from 2022 for 36 of the measures and 2021 for 4 measures (Centers for Disease Control and Prevention, 2024). PFAS concentrations in water may change over time, as may cancer prevalence in a specific zip code, so this could explain some of the variation. Additionally, it is possible that longer-term exposure to high PFAS has a different effect than a single-time high exposure, and cancer may take time to develop after exposure to the chemicals. As a result, including time-based observations in future research may produce different results than those observed in this paper.

In conclusion, while this paper did not find a significant relationship between the concentration of major types of PFAS in water sources and cancer prevalence in the United States, this remains an area that will require future research. As we continue to learn more about the long-term effects of PFAS, its existence in the environment needs to continue to be studied and thoughtfully evaluated. Perhaps future research with access to more data may continue to explore this relationship further.

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